

( 4 )

3. Explain decision variables and constraints.
4. What do you mean by Artificial variable.
5. Explain Infeasible solution.
6. What do you mean by initial basic feasible solution ?
7. How Assignment problem is different from Transportation problem ?
8. Explain different application areas of operation Research.
9. Explain CPM.
10. What are the basic difference between PERT and CPM.

SEM-VI

shilpa

VUG (6)-BCA (C-6001)

2019-22

Full Marks : 70

Time : 3 hours

Answer any five questions.

*The figures in the right-hand margin indicate marks.*

*Candidates are required to give their answers in their own words as far as practicable.*

GROUP—A

Answer any four questions : 10 × 4

1. Explain the scope and methodology of OR. Explain the step involved in the solution of an operation Research problem.
2. Explain General Mathematical model of linear programming problem.

2

3

( Turn Over )



REDMI NOTE 9

VUG(6)-BCA(6001)

ARYAN SIDDHARTHA

Hz-400

( 2 )

$$\begin{array}{r} 125 \\ 24 \\ \hline 500 \\ 250 \\ \hline 3000 \end{array}$$

3. Explain the guidelines on linear programming model formulation with an example.
4. Solve the following LP problem using the simplex method

$$\begin{aligned} \text{Max } Z &= 2x_1 + 4x_2 + 3x_3 + x_4 \\ \text{Subject to constraint s} \end{aligned}$$

$$\begin{aligned} x_1 + 3x_2 + x_4 &\leq 4 \\ 2x_1 + x_2 &\leq 3 \\ x_2 + 4x_3 + x_4 &\leq 3 \\ \text{and } x_1, x_2, x_3, x_4 &\geq 0 \end{aligned}$$

$$\begin{array}{r} 125 \\ 18 \\ \hline 1400 \\ 125 \\ \hline 3950 \end{array}$$

5. Use the penalty (Big-M) method to solve the following LP problem.

$$\begin{aligned} \text{Minimize } Z &= 5x_1 + 3x_2 \\ \text{Subject to the constraints} \end{aligned}$$

$$\begin{aligned} 2x_1 + 4x_2 &\leq 12 \\ 2x_1 + 2x_2 &= 10 \\ 5x_1 + 2x_2 &\geq 10 \\ \text{and } x_1, x_2 &\geq 0 \end{aligned}$$

( 3 )

6. Determine an initial basic feasible solution to the following transportation problem by using (a) NWCM, (b) LCM and (c) VAM.

|        |        | Destinations   |                |                |                |        |
|--------|--------|----------------|----------------|----------------|----------------|--------|
|        |        | D <sub>1</sub> | D <sub>2</sub> | D <sub>3</sub> | D <sub>4</sub> | Supply |
| Source | A      | 11             | 13             | 17             | 14             | 250    |
|        | B      | 16             | 18             | 14             | 10             | 300    |
|        | C      | 21             | 24             | 13             | 10             | 400    |
|        | Demand | 200            | 225            | 275            | 250            |        |

7. Explain Hungarian method to solve an assignment problem.
8. Explain different phases of project management. Also explain procedure to design. Network diagram.

GROUP—B

All questions are compulsory. 3 × 10

1. Explain model in OR.
2. Explain linear programming problem.

REDMI NOTE 9



# VUG (6)–BCA (C-6002)

2019-22

*Full Marks : 70*

*Time : 3 hours*

Answer from **all** the Sections as directed.

*The figures in the right-hand margin  
indicate marks.*

*Candidates are required to give their answers  
in their own words as far as practicable.*

## SECTION–A

Answer any *four* questions :  $10 \times 4$

1. What are management skills ? Explain the types of management skills.
2. What do you mean by management and what are the levels of management ? Discuss.
3. Write down about the contribution of F. W. Taylor.



( 2 )

4. Explain the various functions of management.
5. What is planning ? Explain the steps involved in planning.
6. State and explain the basic steps involved in a typical selection procedure.
7. What are the barriers of effective communication ?
8. What do you mean by 'leadership style' ? Describe the different styles of leadership.

#### SECTION-B

Answer *all* questions : 3 × 10

9. Define Administration.
10. What is scientific management ?
11. What are the objectives of planning ?
12. What are the roles of manager ?

( 3 )

13. What are the needs for communication ?
14. What are the elements of organizing process ?
15. What are the characteristics of control ?
16. Write down about the importance of motivation.
17. What do you mean by coordination ?
18. What are the advantages of decentralization ?

REDMI NOTE 9

ARYAN SIDDHARTHA

( Continued )

VUG(6)-BCA(6002)

Hz-400



14. What is trading account ? What is its uses ?

15. What is profit and loss account ? What is its uses ?

16. What are the different fields that are shown on a balance sheet ?

17. What is the uses of Trial balance in accounts ?

18. What is the purpose of using sub-Journals in accounting ?

2019-22

Full Marks : 70

Time : 3 hours

Answer from all the Sections as directed.

*The figures in the right-hand margin indicate marks.*

*Candidates are required to give their answers in their own words as far as practicable.*

### SECTION-A

Answer any *four* questions : 10 × 4

1. What is accounting standard ? Why is it useful ?
2. Explain classification of account with examples.
3. What are stocks/inventories ? What are the different types of Inventories ?

( Turn Over )

( 2 )

4. Explain Journal and its template and its uses.
5. Explain the uses of Management Information System (MIS) in accounting.
6. What is assets and liabilities ? Explain its types.
7. Explain the different users of accounting information.
8. Pass Journal entries of Siyaram book for the following transactions :

2012

|        |                                                 |          |
|--------|-------------------------------------------------|----------|
| Apr. 1 | Siyaram commenced business with cash            | 2,00,000 |
| 5      | Purchased goods for cash                        | 50,000   |
| 7      | Purchased furniture from Vishal company         | 25,000   |
| 8      | Sold goods for cash                             | 40,000   |
| 10     | Sold goods to Hero Ltd, less 10% Trade discount | 30,000   |
| 14     | Paid Rent                                       | 5,000    |
| 19     | Paid Salary                                     | 10,000   |

( 3 )

2012

|         |                                                      |       |
|---------|------------------------------------------------------|-------|
| Apr. 22 | Received Interest                                    | 500   |
| 24      | Siyaram withdraw cash from business for personal use | 7,000 |
| 27      | Purchased goods from Pawon brother                   | 6,000 |

### SECTION-B

Answer *all* questions :  $3 \times 10$

9. What are the features of accounting ?
10. What are the rules for double entry system ?
11. What is the difference between Financial accounting and Management accounting ?
12. What is Trade discount and Cash discount ?
13. What is book-keeping ? How it is different from accounting ?

REDMI NOTE 9

ARYAN SIDDHARTHA

( Continued )

VUG(6)-BCA(C-6003)

( Turn Over )



VUG (6)— BCA (C-6004)  
Network Security

2019-22

*Full Marks : 70*

*Time : 3 hours*

Answer from both the Groups as directed.

*The figures in the right-hand margin  
indicate marks.*

*Candidates are required to give their answers  
in their own words as far as practicable.*

GROUP—A

Answer any *four* questions :  $10 \times 4$

1. What is attack ? Explain how attack harms computer network.
2. What are the key principles of security ? Also explain the categories of security threats.
3. What is digital signature ? Explain the working process of digital signature.



( 2 )

4. What is the need for security services at transport layer ? Explain four protocol defined by secure socket layer.
5. What is virus ? Explain different types of viruses.
6. Explain RSA algorithms of asymmetric key cryptography.
7. Illustrate in detail about the message authentication code and its requirements.
8. Explain :
  - (a) Block cipher
  - (b) Firewall

GROUP—B

Answer *all* questions :  $10 \times 3$

9. Define risk, vulnerability and threat.
10. What will be the result of an attack on computer network.

( Continued )

( 3 )

11. Explain white and black hat hacker.
12. What is protocol ?
13. What is VPN ?
14. What is phishing ? How does phishing work ?
15. What are the basic objectives of network security ?
16. Explain difference between public key and private key.
17. Why HTTPS should be used instead of HTTP ?
18. What is message integrity and message confidentiality ?

VUG(6)-BCA(C-6004)

Hz-400

REDMI NOTE 9

ARIAN SIDDHARTHA



VUG (6)—BCA (E-6007)  
Mgt. Inf. Sys.

2019-22

*Full Marks : 70*

*Time : 3 hours*

Answer from **both** the Groups as directed.

*The figures in the right-hand margin  
indicate marks.*

*Candidates are required to give their answers  
in their own words as far as practicable.*

GROUP—A

Answer any *four* questions :  $10 \times 4$

1. Write and explain the organization structure ?
2. Write about MIS importance regarding business perspective.

( Turn Over )



( 2 )

3. Discuss about decision making system.
4. Write about criteria of balanced MIS.
5. Explain basic information system.
6. Write the tools/techniques for systematic identification of MIS.
7. Write about DSS.
8. Write and explain the pitfalls in MIS.

GROUP—B

Answer *all* questions :                      3 × 10

9. What is MIS ?
10. What is decision making ?
11. Write the types of information.
12. What is system ?

( 3 )

13. What is the difference between data and information ?
14. Write the types of information.
15. What is Advanced MIS ?
16. What is the need of MIS ?
17. What is Top level management ?
18. Who makes the decision and how ?

( Continued )

VUG(6)-BCA(E-6007)MIS

Hz-400